

How has Mainstream changed?

A Topic Model insight

1. Why studying the mainstream?

To understand the reasons of this work and, eventually, its scientific relevance a premise is necessary.

Economics as a discipline presents very strong social norms and a very hierarchical structure. As a result, it is possible, and quite easy, to identify a small group of journals in which the scholars in the top US departments often publish (Dusansky & Vernon, 1998; Ellison, 2003; Heck & Zaleski, 2006; E. H. Kim et al., 2006, 2009; Card & DellaVigna, 2013; Hamermesh, 2013). Those are the same journals that are able to assure a scholar the tenure in the very same departments, by publishing one of their papers (Heckman & Moktan, 2020).

In other words, there is a group of journals (the Top5 or the Blue Ribbon Eight) which are liked (and managed) by those who are generally liked (and cited and funded) by the discipline.

This concentration of power and, as a consequence, prestige has inevitably created a notion of prestige in every aspect of the discipline: there are more prestigious theories, more prestigious research topics, more prestigious departments and so on.

But it happens quickly, in the underfunded academia, that the most prestigious alternatives become the only alternatives, inducing conformism and, as a matter of necessity, excluding less prestigious theories from scientific debate.

The prestigious economics is, politely, known as *mainstream economics* or, in a more colourful way, as *orthodox economics*. The idea of an orthodoxy, and so of some competing heterodoxies, move the analysis to the semantic field of religion and faith. And it is intentional.

As many, mostly heterodox, scholars have pointed out, many of the hypotheses which support the mainstream economics have to be accepted by faith, since there is not real-world evidence of their realism¹. The consequence is that to pursue a career in economics and be able to get a prestigious (and thus well-funded) position, a profession of faith is required of every (young) scholar².

The last piece of the puzzle required to justify this work is the role of economics in society.

Economics is a social science which influences the kind of policies realized by the governments. It provides theoretical justification, it carries out the forecasts on the real-world effects of policies,

¹Some orthodox scholars address this criticism by not considering realism as necessary in a scientific theory, but discussing this would be a too long digression.

²An example of this kind of “religious” reasoning is proposed by Galbraith: “Accepted in reputable market orthodoxy is, as noted, the inherent perfection of the market. The market can reflect contrived or frivolous wants; it can be subject to monopoly, imperfect competition, or errors of information, but, apart from these, it is intrinsically perfect. Yet clearly the speculative episode, with increases provoking increases, is within the market itself. And so is the culminating crash. Such a thought being theologically unacceptable, it is necessary to search for external influences—in more recent times, the downturn in the summer of 1929, the budget deficit of the 1980s, and the “market mechanisms” that brought the crash of 1987. In the absence of these factors, the market presumably would have remained high and gone on up or declined gently without inflicting pain. In such fashion, the market can be held guiltless as regards inherently compelled error. There is nothing in economic life so willfully misunderstood as the great speculative episode.” (Galbraith, 1994)

it influences the public debate providing the words to convey different ideas of the economy (and so of the society), helping to move Overton's window.

In other words, Economics is a scientific discipline with a strong impact on society, and the social and political implications of any economic theory must not be overlooked³.

So, critically studying the most widespread economics is necessary to understand which conceptual framework the discipline is providing to the society to understand the reality, to unveil which cultural and political programme it is (more or less consciously) carried out by the discipline.

Nevertheless, this work only tries to describe mainstream economics, without arguing on the social consequences of the features highlighted. Which is still a necessary first step.

2. The interpretative hypotheses

I am not the first who tries to describe mainstream economics, and neither the first to do it quantitatively (I review some previous attempts in the next section).

Therefore, I can climb up on the shoulders of other scholars to be guided to what I can look for and which interpretative frameworks I can use.

Two qualitative theories will be the core pieces to guide this work: the *empirical turn* hypothesis and the *mainstream pluralism* hypothesis.

The first lens with which to observe the mainstream economics is the progressive shift toward empirical research (Backhouse & Biddle, 2000; Backhouse & Cherrier, 2016, 2017; Cherrier & Svorenčík, 2018). Neoclassical economics, which is the theoretical framework on which mainstream economics is based since the seventies, built its fortune on the ability to provide clear models expressed using a very formal flavour of mathematics, mostly calculus-based. In more recent times instead, in part as a reaction to the credibility crisis arose after the financial crises of the first decade of the XXI century, econometrics is becoming the standard mathematical tool of analysis and theory-driven models are being replaced by data-driven models. This empirical turn in the scientific practices of the discipline is moving researchers away from the comprehensive economic models of the past, towards a perilous land in which the only theory is the way in which data are analysed, and every economic insight can come only from data itself.

The second lens is the mainstream pluralism hypothesis (Davis, 2006; Cedrini & Fontana, 2018; Davis, 2019a, 2019b). It has been observed that the domain of research of economics is widening and the number of different computational and mathematical tools used is increasing⁴. These two tendencies have allowed the emergence of many new fields in the discipline, which are very focused on a particular topic (like environmental or health economics) or on a particular method (like economic complexity or behavioural economics). Researchers who are specialized in one of these fields rarely engage with the other niches, creating an archipelago of unconnected islands and an appearance of pluralism in economics. The mainstream pluralism hypothesis argues that this process of specialization is not developing pluralism as historically intended in economics (i.e. as a plurality of competing ontological views of the economy and the economics), but rather it is creating a plurality of loosely connected fields. These share a common ancestral set of hypothesis (the neoclassical one) and relax some of them to be able to tackle specific

³The conceptual framework provided by Keynes' theories is necessary to historically and politically understand Roosevelt and the European social state of the 50s. Similarly, Margaret Thatcher's and Ronald Reagan's economic reforms would have never been realized without the theoretical background provided by Friedman.

⁴I want to highlight two factors that have probably helped this shift. The first are the new data-driven methodologies, which are able to deal with a wider range of hypotheses than the neoclassical calculus-based approach (Cherrier & Svorenčík, 2018). The second one is the return from the imperial conquests of other social sciences (Fourcade et al., 2015; Marchionatti & Cedrini, 2017), which has widened the domains of research and introduced new methodologies, like the laboratory experiments that behavioural economists have learnt from psychology.

problems, putting them in continuity rather than in contrast with neoclassical economics⁵.

In the quantitative exercise that constitutes the core of this paper, I try to find some evidences in favour of these two hypotheses.

3. Distant Reading and quantitative methods in the History of Economic Thought

A common problem in the study of the history of ideas (of which the history of economic thought is surely part) is the impossibility to accurately read a big number of textual documents in a reasonable time. A first, and often adopted, solution is to select a sample of relevant works, like the books of a single author or a handful of representative masterpieces of a period, and analyse only them. An alternative approach is what Franco Moretti called Distant Reading (Moretti, 2013): instead of carefully (close) reading few works, a big number of documents are summarily analysed using statistical methods and the aid of computers. Different features of a text can be analysed, but generally the focus is on the textual content of the document (recovered by a digital version or through OCRing) or some metadata (title, citations, authors' list).

The kind of data used and the research questions tackled move Distant Reading close to bibliometrics and text mining studies. The fil rouge which ties together Distant Reading studies is their motivation: to overcome with quantitative techniques the impossibility to (close) read thousands of documents to reconstruct the history of some idea.

The history of neoclassical economics as mainstream economics covers at least thirty years and many journals. Moreover, its influence has been systemic, it has shaped the social norms of the discipline as a whole, and sampling only a handful of “representative masterpieces” (assuming the existence of a good enough criterion to select them) would not allow to return all the details I’m looking for.

To get a broader picture, other scholars already applied quantitative methods to the history of economic thought in order to represent the evolution of the discipline in the last half-century.

In 2018, an entire special issue of the *Journal of Economic Methodology* was dedicated to the topic (Cherrier & Svorenčík, 2018; Edwards et al., 2018), but the trail goes way back in the past (Backhouse et al., 1997). The following is a brief review of some interesting works which address similar questions to mine.

Using JEL codes⁶, Kelly and Bruestle (2011) observe a reduction of microeconomics and macroeconomics papers and an increasing of finance and development economics papers. Moreover, the top journals over-represent microeconomics, mathematical and quantitative methods and labour economics.

Hamermesh (2013) shows that the quota of empirical articles in some top journals is increasing.

Card and DellaVigna (2013) observe a bias toward theoretical microeconomics papers (and to a lesser degree applied microeconomics and macroeconomics ones) in top journals, which is reducing in the last years as empirical papers become more frequent.

Claveau and Gingras (2016) clusterise the citation network to highlight seven subfields which are present during the entire studied period (1956-2014): econometrics, financial economics, development and trade economics, industrial economics and game theory (the largest subfield after 2000), history of economics and macroeconomic policies, macroeconomics and monetary policies, labour economics. In addition, they observe some clusters only at the beginning of the sequence, which are not preserved in subsequent years, and one cluster (environmental

⁵In this sense they can be viewed as subfields of a wider “post-neoclassical” economics, which aims to improve (or save) the neoclassical approach and legacy, rather than rebuild the discipline on completely different assumptions like the heterodoxies aim to do.

⁶The *Journal of Economic Literature* (JEL) codes are a widely used taxonomy system to classify economics subfields, but their reliability is discussed, since their use appears to be highly subjective (Cherrier, 2017; Kosnik, 2018).

115 economics) which appears toward the end of the studied period. The increase in published papers makes the recognition of clusters more difficult at the end of the series, and it is possible that some clusters went undetected in the most recent years. Finally, they observe that it is typical for economics scholars to publish in more than one subfield.

120 Angrist et al. (2017) find that the share of microeconomics (mostly theoretical) papers is steadily increasing, as well as the share of development economics ones, while typical applied microeconomics fields (like labour economics or industrial organization) see their shares reducing over time. Moreover, they describe the increase of empirical papers as a methodological shift inside the subfields, rather than a change in research topics.

125 Ambrosino et al. (2018) explore the whole JSTOR database on a decade-per-decade basis, using a Topic Model, going deep in describing the possibilities of this kind of approach.

Fontana et al. (2019) observe an increasing usage of treatment effects models at the expense of time series, a growth of development economics, game theory and education economics and a decline of industrial economics and macroeconomics. This study uses LDA on the papers' full-text.

130 Montesinos and Brice (2019) measure the most frequent words in the title of the most cited papers of the last seventy years, highlighting a shift from "theory" to "evidence" (and so empirical research).

Except for Ambrosino et al. (2018), all the studies listed focus on a small subset of the economic journals, usually the "Top5"⁷ or the "Blue Ribbon Eight"⁸.

135 4. The choice of the data

Going back to the aim of this paper, the two questions to be answered pertain the evolution of economics at least since the seventies. But, ideally, recovering data since the post-war period would allow observing both a pre- and a post-neoclassical period.

140 Moreover, the focus is on mainstream economics, which allows a strong hypothesis in choosing the data: the journals perceived by the research community as the top journals are a representative sample of what is mainstream in the community, i.e. what is considered common knowledge, high quality research and a signal of the path to follow.

145 To select the top journals one can rely on the "common knowledge" represented by the Blue Ribbon Eight, the rankings published throughout the history of the discipline or looking at some bibliometric indicator. The choice is obviously restricted by the availability of different sources of data.

There are three data types available: citation networks, articles' metadata (particularly JEL codes) and articles' text (or their abstracts).

150 I exclude the JEL codes because they are highly subjective and can change multiple times during the editorial process (Kosnik, 2018). Furthermore, they are used to signal the (desire of) belonging to a scientific community which can be interested in reading the paper. They are, indeed, more a way to observe by whom the scholars desire to be read, and so the relations inside the discipline rather than the evolution of the ideas.

155 Similar reasoning can be done with the citation network. Especially in more recent times, citations have become a currency which buys funding and (tenured) positions. According to Goodhart's law, they have become a poor metrics to measure the relevance of a paper, since the scholars started to play the game.

This reasoning leads me toward the use of the text of the articles. This strategy has an obvious downside (it is more difficult to analyse textual documents than a network or a taxonomy), but

⁷ *The American Economic Review, Econometrica, Journal of Political Economy, Quarterly Journal of Economics and Review of Economic Studies* (Heckman & Moktan, 2020).

⁸ The Top5 plus *International Economic Review, Journal of Economic Theory* and *Review of Economics and Statistics* (Dusansky & Vernon, 1998).

it reduces some disadvantages of the two other ones. To be honest, the text of a document is still influenced by the community in which a researcher puts himself (the same idea is expressed differently by scholars of different schools of thought) but it is harder to be gamed (and maybe it is not a real downside being able to distinguish similar but rival communities). Finally, the selection bias of the authors in writing the abstracts can be avoided by using the full text of the articles, available through JSTOR.

This choice raises a problem: not every journal is indexed on JSTOR. Particularly, of the Blue Ribbon Eight the *Journal of Economic Theory* is missing. I believe this absence not to be significant, since the Top5 are still available, it (and the *International Economic Review*) has slightly worse bibliometric indicators than the other six, and the sample is still pretty large⁹.

The JSTOR database is particularly extended in time, going back to the early XX century. As said before, this allows to include in the study a period before the rise of neoclassical economics and mainstream economics, the late Keynesianism of the fifties. On the other hand, the use of the Top5 as a proxy can suffer from recentism, since their prestige has consolidated during time.

Looking at some older journals' ranking (Coats, 1971; Billings & Viksnins, 1972; Moore, 1972; Hawkins et al., 1973; Liebowitz & Palmer, 1984; Malouin & -Francois Outreville, 1987) two other journals appears to be quite unanimously considered prestigious at the time: *Economica* and *The Economic Journal*. Luckily, both are available in JSTOR¹⁰.

The resulting sample is composed by: the Top5 (*The American Economic Review*, *Econometrica*, *Journal of Political Economy*, *Quarterly Journal of Economics* and *Review of Economic Studies*), one additional Blue Ribbon (*Review of Economics and Statistics*) and two "old glories" (*Economica* and *The Economic Journal*).

Finally, only the research article published between 1946 (the first year after the WWII) and 2016 (the last year for which the full-texts of the articles was available for the eight journals in the sample at the time of starting the project) are considered¹¹.

5. Strategies of analysis

The research question of this work (if mainstream economics is going through an empirical shift and a pattern of specialization) is about the topics dealt with in mainstream economics (using a sample of top journals as proxy). But classifying textual documents based on their topics is not an easy task. The common assumption is that if two papers use the same words, they probably are about the same topic. Even with this assumption, a set of lists of words (which is a more formal representation of a text) is still something without a clear and manageable mathematical abstraction.

The strong hypothesis which is commonly used in this context is to assume that the information contained in the order of the words is negligible with regard to the information conveyed by the frequency of the words. This is known as Bag-of-Words (BoW) approximation, because

⁹As shown in Table 1, the Top5 plus the *Review of Economics and Statistics* are the only six generalist journals in the Top20 of each of the three different bibliometric indicators computed by SCImago (average of the 1999-2021 period).

¹⁰There is a strong third candidate to be added to the sample: *The Bell Journal of Economics*, later become *The RAND Journal of Economics*. It was not included because I, the author, did not know that the journal had changed its name during its history and, as a consequence, I decided to exclude it because its publication period (as the Bell Journal of Economics) does not cover all the chosen time-span. By the time I discovered the error the dataset was already consolidated. This is a brilliant example of path dependency in research and of the need of a strong domain knowledge to do quantitative studies. Nevertheless, it is the less strong candidate of the three.

¹¹The date were downloaded from Constellate on the 19/09/2022. A later attempt to renew the dataset has been made on the 21/09/2026, but the constellate service were shut down and the new JSTOR dataset does not include two top5 journals (*Quarterly Journal of Economics* and *Review of Economic Studies*). For this reason the older dataset has been used.

Table 1: The 20 journals with the highest average indicators in the period 1999-2021 for three different bibliometric indicators. Journals from the subject “Economics, Econometrics and Finance”. Journals containing the words “Financ”, “Marketing”, “Account”, “Business”, “Entrepreneur” or “Consumer” are excluded. Sourced from <https://www.scimagojr.com/journalrank.php>.

SCImago Journal Rank		H-index		2-year Impact Factor	
Quarterly Journal of Economics	33.07	American Economic Review	399	Quarterly Journal of Economics	17.30
Review of Economic Studies	23.62	Quarterly Journal of Economics	336	Journal of Innovation and Knowledge	17.04
American Economic Review	23.49	Ecological Economics	279	Energy Economics	14.60
Econometrica	19.93	International Journal of Production Economics	264	Journal of Economic Literature	13.15
Annual Review of Economics	17.77	Journal of Economic Perspectives	253	Annual Review of Economics	13.14
Journal of Economic Literature	16.87	Energy Economics	252	International Journal of Production Economics	12.77
Journal of Political Economy	16.25	World Development	249	Resources, Conservation and Recycling	12.17
Review of Economics and Statistics	8.83	Resources, Conservation and Recycling	245	Internet Research	11.71
Journal of Economic Perspectives	8.18	Econometrica	242	Review of Economic Studies	11.42
Journal of Labor Economics	6.77	Journal of Political Economy	235	Electronic Markets	11.32
Journal of Econometrics	6.58	Review of Economics and Statistics	219	Journal of Economic Surveys	10.65
Journal of Monetary Economics	6.26	Economic Journal	205	Journal of Economic Perspectives	10.62
Econometrics Journal	6.12	Journal of Econometrics	205	Economic Analysis and Policy	10.58
Journal of Human Resources	6.09	Journal of Economic Literature	199	Economic Geography	10.49
Economic Journal	5.18	Journal of Development Economics	196	Resources, Conservation and Recycling Advances	9.87
Journal of International Economics	4.84	Journal of Public Economics	192	American Economic Review	9.55
RAND Journal of Economics	4.62	Review of Economic Studies	184	Resources Policy	9.31
Quantitative Economics	4.59	Journal of International Economics	179	Cleaner and Responsible Consumption	8.26
Brookings Papers on Economic Activity	4.45	Journal of Environmental Economics and Management	168	Telecommunications Policy	8.15
Journal of Public Economics	3.60	Journal of Monetary Economics	165	Annual Review of Resource Economics	7.98

it is like if the words composing every document would be put in a bag and shaken until reconstructing the order becoming impossible. This hypothesis is used also in this study.

Using the BoW approximation the corpus can be represented as a matrix with the documents on the rows, the different words on the columns, and the frequencies as entries. The same matrix represents also a bipartite graph and a linear space of documents spanned by the words.

With a mathematical object available, it is necessary to manipulate it to obtain some answers. Two alternatives are available: formulating the broad research question in a very precise one, for which it is easy to measure a proxy, or using an algorithm to transform the representation of our dataset and then take some measures on the new representation. The second path is followed.

Before moving into the cleaning and the analysis of the data, it remains to discuss which algorithms can be used to obtain a new, and more useful, representation of the data. A representation able to highlight semantic similarities (i.e. the topic) among documents is known in literature as Topic Model.

Common strategies in literature include inferring a latent stochastic process (Hofmann, 1999; Blei et al., 2003; Griffiths & Steyvers, 2004; Teh et al., 2005), using clustering techniques on the bipartite network (Gerlach et al., 2018) or on the vector space (Angelov, 2020; Grootendorst, 2022), reducing the dimensionality of the matrix (J. Kim & Park, 2008), or a combination of the previous (Lancichinetti et al., 2015)¹².

The oldest algorithms, and among them the very popular Latent Dirichlet Allocation (LDA Blei et al., 2003), require the scholars to choose a priori the number of topics to be recognized, or equivalently in how many groups to divide the documents. This is clearly a strong downside of these algorithms, and, as far as I know, in literature there is not any attempt to characterize the stability of these algorithms with regard to changes in the number of topics. Moreover, most of these algorithms show little stability for different random seeds and initial conditions.

To overcome the limitation of not having to set a priori the number of topics, two roads have been explored: improving LDA (Teh et al., 2005; Lancichinetti et al., 2015) or going for a totally different paths. Neither HDP (Teh et al., 2005) nor TopicMapping (Lancichinetti et al., 2015) achieve the result effectively: preliminary investigation shows that, for a dataset which is not clearly divided, HDP predicts a number of topics very close to the minimum number of topics provided by the researcher, while TopicMapping collapses in few (even one or two) very big and uninformative clusters. I cannot exclude that different choices in preprocessing the dataset, which is –as I will discuss later– a very arbitrary process, can resolve those problems.

A different path is to use a combination of a dimensionality reduction technique, like UMAP (McInnes et al., 2020), followed by a clustering algorithm, like HDBSCAN (Campello et al., 2013; McInnes et al., 2017), to classify the document. This strategy should show greater consistency across different random seeds but also change the type of output obtained. LDA returns the probability for each document to belong to each topic. HDBSCAN, instead, associates each document to at most one topic, leaving some documents unclassified. Regarding the number of topics, the researcher is able to set the minimum number of documents for each topic, allowing to resolve the model at different scales. Finally, this kind of workflow is often preceded by an embedding algorithm which provides a change-of-basis function from the word-spanned vector space to an abstract one, determined with machine learning techniques (Angelov, 2020; Grootendorst, 2022). The presence of unclassified documents, and a personal preference to not use embedding techniques, has put me on another different path.

The strategy used in this work relies on the hierarchical stochastic block model algorithm (hSBM, Gerlach et al., 2018; Peixoto, 2019), which aims to clusterize each partition of the bipartite graph, at different scales. It yields a collection of models, each one a zoomed version of the previous, in which each word is assigned to a topic and each document is assigned to a group. For those used to LDA this is an important difference: LDA infers a single latent layer

¹²An attempt of comparison is available online at <https://github.com/TnTo/TopicModelBenchmark/>

Table 2: Number of documents, unique words and tokens in the dataset at three different stages of the cleaning process

	Documents	Words	Tokens
Original dataset	81309	10323373	438722824
After first cleaning	35513	2040481	178900475
Final dataset	32985	32520	38041150

(i.e. Words—Topic—Documents), while hSBM infers two of them (Words—Topics—Groups—Documents). Moreover, hSBM provides both a hard-clustering version (like UMAP-HDBSCAN in which each document or word belongs to only one cluster) and a soft-clustering version (like LDA in which the membership is probabilistic). The problem of this algorithm is its computational cost which is prohibitive for the soft-clustering version and still elevated for the hard-clustering one¹³.

The final choice is anyway hSBM, mostly because it is the algorithm that requires the researcher to input the smallest number of parameters, it allows zooming into each group to analyse its composition (i.e. of which subgroups is composed) and it provides an unsupervised method to choose among the results of different initial conditions.

6. Corpus cleaning

The starting point of this analysis¹⁴ is a dataset downloaded from Constellate, a web-based software to analyse papers in JSTOR database, which includes all the available documents published in the seven available Blue Ribbon Journals, *Economica*, the *Economic Journal* and the *Bell Journal of Economics*¹⁵.

First, the dataset is converted from the original jsonl format to a SQLite database. The resulting dataset is the “Original dataset” in table 2.

Then the dataset goes through a first cleaning step. The articles published in one of the eight journals of the sample between 1946 and 2016 are selected. Of these, only those with at least one author listed (this heuristic discards mostly editorial articles, notes, comments and similar non-research articles), indexed on JSTOR, classified as “research-article” and with the unigrams count (i.e. the list of the words in each article with their frequency), are kept. Furthermore, non-alphabetical characters are removed from each word listed, which is then lowercased and stemmed (with the NLTK Snowball stemmer). Then words composed of one or two characters are discarded. The result of this step is the second row of table 2. 55% of the original documents are discarded in this way and the number of unique words is reduced by 80%.

This intermediate version of the corpus discards all the obvious non-inclusion from the dataset. It discards over-downloaded articles (selecting by journal and year), wrong type of contents (notes, indexes, editorials...) and a first group of non-informative words. Particularly, it modifies

¹³The final version of the dataset requires more than 10GB of RAM to be processed and more than 10 hours on my old personal laptop. One cause of this is the single-threaded nature of the algorithm. A previous version of the dataset, with a smaller number of words removed, required more than 20GB of RAM. The dataset with the documents’ metadata is a 300MB SQLite file.

¹⁴The steps of the analysis can be followed and reproduced using the files `main.py` and `mainstream.py` at <https://github.com/TnTo/mainstream/>.

¹⁵The Constellate IDs of the dataset downloaded are: `31bcd322-032f-2ba5-37c4-c2ca96952ebf` (7 Blue Ribbons - 1900-1954), `1a0cd895-717b-dcc2-940d-a4d7ed069aea` (7 Blue Ribbons - 1955-1984), `03aa84ad-ec5e-290c-69e7-721535714c9e` (7Blue Ribbons - 1985-2022) (Downloaded 19/09/2022), `4788f182-8ec3-8eb8-ac5b-a1b318b03dbb` (*Economica* and *Economic Journal*), `c55cf30b-737b-9a83-3325-8d82a71bd8e6` (*Bell Journal of Economics*) (Downloaded 08/10/2022). Constellate website was off-lined in July 2025.

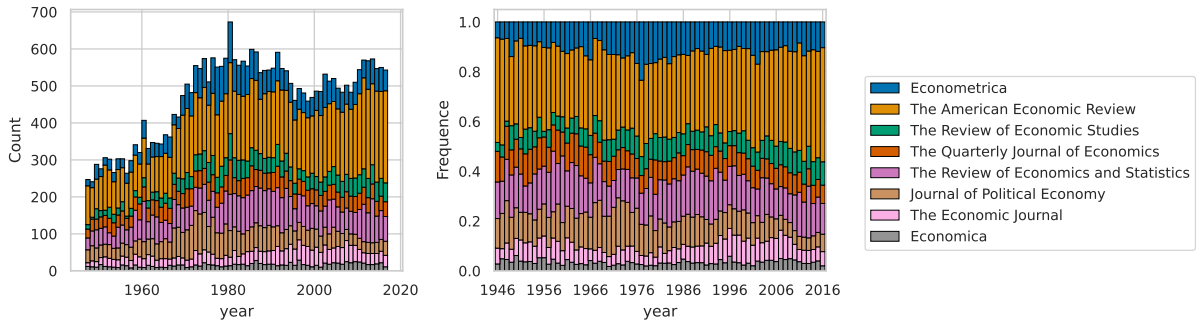


Figure 1: These two plots represent the number of documents in the dataset divided by year and journal. On the left, the absolute number is reported, while on the right the same data are reported normalized by year.

OCR errors removing number and typographic characters and discards very short words which are either non-informative on their own (I, a, of, on, it, ...) or too generic after stemming.

The only non-obvious choice in this step is whether to stem or not to stem. It is safe to assume that the semantic difference among, for example, stutter, stutters and stutterer is not relevant in describing the broad topic covered. In this way we are able to remove some noise from the input of the algorithm, selecting (using some previous linguistic knowledge) only the relevant part of the information. A second argument is more practical: stemming significantly reduces the dimension of the documents-words matrix, allowing the topic model algorithm to run faster with less resources. But stemming is not the only way to achieve this, and similar results can be obtained also using lemmatization or embedding.

The second cleaning step is more arbitrary. First, only the words (that at this point have already been lowercased and stemmed) which appear in at least 35 documents (approximately 1 every 1000) and at most in 9000 documents (approximately one fourth of the corpus) are kept. This lower and upper bounds are determined by the desired type of results, in other words how many groups are expected. The answer is ten to twenty, because less than ten groups would be too broad and archetypal (macroeconomics, microeconomics, econometrics and few others) to capture any change, while more than twenty could be too detailed to be easily distinguished and interpreted. Following this reasoning, a word which compares in over a quarter of the corpus characterizes a topic too broad to be of interest (if not the discipline as whole), while a word which compares in less than a document every one thousand brings a shade of meaning too specific to be of interest at the scale of interest (or more probably it is an OCR error). Then, a small set of `\LaTeX`command¹⁶ are removed from the list of words, because they signal not a topic but a way of typesetting and digitalizing (and so, probably, a journal and a period). Finally, only the documents with at least 1500 tokens (i.e. total words) after the first cleaning step are kept. Those are documents longer than three or four pages, and it is another heuristic to remove errata corriges, short notes and similar documents.

At the end the final dataset, on which the topic model is inferred, is composed by slightly more than thirty thousands documents (93% of the intermediate dataset), thirty thousands unique words (2%) and thirty-eight million tokens (21%).

Figure 1 shows the absolute and relative composition of the final dataset disaggregated by journal and year. The most represented journal is the AER, and it is possible to observe the slowdown in publication described by Ellison (2003) and Card and DellaVigna (2013).

¹⁶`\landscap, \normalfont, \usepackageamsbsi, \renewcommandsfdefaultwncys, \declaremaths, \usepackagestmaryrd, \usepackageetextcomp, \renewcommandrmdefaultwncyr, \newcommandcyr, \usepackageototfontenc, \pagestyleempti, \renewcommandencodingdefaulttot, \declaretextfontcommandtextcyr, \documentclassaastex, \usepackageportlandxspac, \usepackagepifont, \usepackageamssymb, \selectfont, \usepackageamsmathamsxtra, \usepackagemathrsfs, \usepackagebm, \usepackageamsfont, \begindocu, \enddocu, \iin, \frac.`

Table 3: Number of groups (both words and documents) in each level of the hierarchy for each random seed

level seed	0	1	2	3	4
1000	1369	185	32	5	2
1001	1207	179	26	5	2
1002	1494	193	34	6	2
1003	4288	479	64	11	2
1004	1508	312	57	8	2

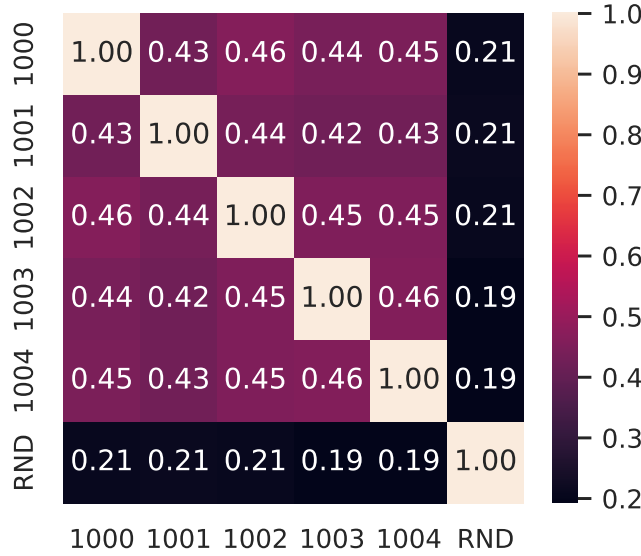


Figure 2: Normalized mutual information for the five level 2 partitions and a random control partition (RND)

7. Model selection and validation

The hSBM algorithm is sensitive to the initial conditions, provided as random seed. For this reason five different random seeds are given to the algorithm and the five outputs are compared.

Table 3 shows the instability of the model: while the number of levels in the hierarchy is the same for each random seed, the number of groups (composed by either documents or words) at each level varies. Furthermore, the magnitude of the number of levels is the same independently of the random seed (and so they describe the model at the same scale).

For the reasons stated above, the level of the hierarchy at the scale of interest is the level 2.

To verify that the similarity is not only in the magnitude but also in the composition of the groups a null-model is generated. Focusing on the level of interest, the average number of words-groups and documents-groups are computed. Then, words (documents) are randomly divided in such number of groups. The resulting partitions (the null-model and the five topic models) are compared using the Normalized Mutual Information. As shown in figure 2 the mutual information between the five topic models is higher (.42-.46) than the mutual information between one of the topic models and the control partition (.19-.21). This is a strong hint that the five models recognize some kind of structure (the topics) and their similarity is not a coincidence.

At this point it is necessary to chose one of the five models to analyse in detail. The algorithm chosen (hSBM) provides a measure of the goodness-of-fit for both the complete hierarchy and each level. The metric is the entropy of the partition inferred, which, qualitatively, expresses

Table 4: The table reports the entropy of the inferred partition for the whole model and for the level of interest, per each random seed (lower is better)

seed	Model Entropy	Level 3 Entropy
1000	67317377.964890	60979.976111
1001	67351390.642325	57600.056917
1002	67462251.909423	65728.927821
1003	66917015.060912	291062.829686
1004	67350715.542653	151643.293315

how much information is needed to describe the model given the partition. So, the lower the entropy the more informative on itself is the partition.

From table 4, the model with random seed 1003 is the one with the lowest entropy of the hierarchy, while the model 1001 is the one with the lowest entropy for the level we are interested in (and it is also the one with the smallest amount of groups in that level, see table 3). Since in this work I don't exploit the presence of a full hierarchy, but I analyse only a single level, the choice of the model is driven by the entropy of the level, and so the model 1001 is chosen, and it is the only one considered in the following section¹⁷.

8. Interpreting the model

The chosen partition is composed by 11 groups of documents and 15 topics (or groups of words). To interpret the model it is useful to give a name to each topic and group.

To label the groups the occurrences of each word in each document of the group are summed together, then the most frequent words in the group are used to get a general sense of the group (table 13). Moreover, considering this sum as a single document, the most similar documents (using the cosine similarity) to the group as whole are also listed (see table 9 in appendix).

For the topics (the groups composed by words), instead, the most frequent words in the corpus are listed for each topic in table 6. *Stop Words* is the topic which regroups words with different meanings and which do not bring a semantic insight.

Once both groups and topics are labelled, it is possible to express a group as a combination of topics, as in table 7¹⁸.

This is, obviously, only one possible strategy to label topics and groups, but it is the most commonly used, since it does not require additional data.

Finally, the evolution of groups and topics in time is reported in figure 9 and 4 (the plot reports the graph smoothed with a 10-years moving average).

8.1. The empirical turn

Looking at the groups over time plot (figure 9) some very noticeable trends appear.

First the group labelled *Trade* (which includes *war* and *union* among the most frequent words) is by far the most prevalent group at the beginning of the period and rapidly reduces its presence in the following years. This can be linked (but a more careful look is needed) to the progressive disappearing of the (original) Keynesianism and the classical political economy, in addition to the loss of interest in planned economies.

¹⁷For the other models see the appendices.

¹⁸This representation suggests an interesting thought. Science evolves also by recombining older ideas, and we are in fact hypothesizing, by using this model, that the same group of words has a different meaning if used in different contexts. Algorithms, like LDA, with a single intermediate layer provide a similar insight allowing each word to belong to different topics, and more than one topic to be present in a single document.

Table 5: The documents-group and the twenty most frequent words in each of them. Seed 1001

Applied Micro	household, educ, hous, black, union, hour, citi, crime, panel, job, dummi, white, women, instrument, match, children, yes, health, student, femal
Econometrics	asymptot, matrix, lemma, likelihood, moment, converg, covari, densiti, instrument, finit, null, uniform, root, simul, identif, nonparametr, sequenc, nonlinear, stochast, bootstrap
Game Theory - Theory	game, player, payoff, lemma, belief, equilibria, seller, buyer, bargain, signal, bid, nash, princip, finit, experiment, auction, learn, sequenc, axiom, pair
Game Theory – Auctions	auction, bid, bidder, match, player, game, student, insur, buyer, signal, payoff, des, seller, les, lemma, investor, educ, equilibria, win, pair
Game Theory – Elections	vote, voter, parti, game, elect, entri, buyer, payoff, job, signal, candid, equilibria, seller, match, unemploy, incumb, commit, household, bid, belief
Industrial Organization	patent, manufactur, plant, compani, land, agricultur, citi, hous, entri, household, innov, foreign, locat, export, farm, capac, dummi, advertis, credit, corpor
Labour	job, educ, unemploy, household, women, hour, health, occup, men, children, skill, panel, male, dummi, black, birth, insur, white, week, train
Macro 1	inflat, shock, export, tariff, cycl, foreign, domest, hous, deposit, credit, forecast, unemploy, debt, household, nomin, inventori, bond, liquid, spend, loan
Macro 2	inflat, export, shock, forecast, foreign, unemploy, bond, debt, domest, volatil, labour, cycl, manufactur, innov, nomin, credit, portfolio, currenc, equiti, dividend
Macro Econometrics	inflat, shock, unemploy, path, lemma, debt, equilibria, nomin, matrix, labour, liquid, convex, trader, converg, job, sequenc, foreign, investor, domest, household
Trade	household, educ, student, export, children, union, agricultur, foreign, health, parent, women, insur, child, score, land, war, credit, men, hous, unemploy

The centre of the period (the seventies and the eighties) are characterized by the peak of Macroeconomics and later Game Theory (divided in more topics), which is reasonable to trace it back to the golden period of neo-Keynesian and neoclassical economics.

Some groups show a growth from the beginning to the end of the time span: among them *Labour* and *Applied Microeconomics*.

Similar considerations emerge, with less clarity, by analysing the topics (figure 4). Particularly the topics labelled *Applied Micro* shows the strongest growth in the period.

In appendix the same graph and word-list for the groups of documents obtained with the other seeds are reported: similar dynamics appear (particularly in seed 1000) with an early classical economics, a middle mathematize Macroeconomics and a late Applied Micro/Game Theory cluster.

Those observations allow answering positively to the question “Has there been an empirical turn in economics and is it possible to measure it?” (which, to be honest, has been already done in literature with other methods).

8.2. Mainstream pluralism?

The way the model has been presented tells very little about the other focal point of the study: the mainstream pluralism hypothesis. The empirical turn described above has been described as a condition that facilitates fragmentation and specialization (Backhouse & Cherrier, 2017; Cedrini & Fontana, 2018), but on its own is only a clue.

Table 6: The twelve topics (words-group) and the twenty most frequent words for each of them

Anti-Trust	cartel, herd, franchis, cascadi, barrel, boycott, fad, ivo, sushil, conspir, lusion
Applied Micro 1	household, shock, hous, panel, insur, hour, cycl, instrument, entri, skill, matrix, buyer, innov, path, converg, budget, home, vote, seller, extern
Applied Micro 2	educ, job, student, dummi, health, children, parent, score, status, child, occup, male, census, femal, birth, cohort, communiti, retir, counti, immigr
Banking	borrow, bond, capac, corpor, liquid, portfolio, deposit, inventori, dividend, durabl, oil, item, trader, valuat, billion, default, entrepreneur, gold, gnp, energi
Credit	credit, debt, loan, simul, post, volatil, residu, target, regim, persist, flexibil, record, nonlinear, frequenc, identif, crosssect, anticip, largest, six, hypothes
Crisis	crisi, crise, destin, ricardo, imf, grilich, fish, jorgenson, mcfadden, translog, theil, dualiti, sargan, zellner, nerlov, diseconomi, butter, diewert, pollak, berndt
Development	women, land, farm, gdp, network, fertil, season, merger, farmer, leisur, gender, india, villag, vehicl, pollut, drug, ethnic, brand, diffus, border
Elections	profil, incumb, coalit, card, peer, decentr, grossman, corrupt, prize, recruit, unanim, bribe, bureaucrat, alesina, utilitarian, fehr, wtp, persson, camer, veto
Game Theory 1	game, payoff, bid, auction, bidder, nash, messag, oppon, node, noncoop, weiss, roth, fudenberg, krep, twosid, crawford, rationaliz, oneshot, vickrey, firstpric
Game Theory 2	player, subgam, maskin, sender, myerson, urn, strategyproof, harsanyi, segal, palfrey, subgameperfect, sobel, pesendorf, purestrategi, selten, undomin, xpi, acycl, abreu, supergam
Industrial Organization	union, match, citi, plant, locat, day, million, protect, food, patent, distanc, urban, advertis, aid, ownership, car, agenc, legal, conflict, minor
Labour	unemploy, export, inflat, foreign, labour, domest, manufactur, spend, forecast, tariff, agricultur, war, nomin, fiscal, cash, currenc, expans, perman, fluctuat, transport
Maths	lemma, equilibria, bargain, learn, belief, signal, investor, equiti, stochast, option, convex, indiffer, behaviour, mathemat, she, strateg, concav, switch, ineffici, pareto
Names	frenkel, overshoot, chipman, razin, morishima, carlson, indecompos, goodhart, dmt, feig, quirk, georgescuroegen, cofactor, nikaido, dmr, mcmanus, charn, tanzi, osaka, makin
Stopwords	yes, longrun, her, top, access, length, static, mix, format, physic, upper, max, symmetr, repeat, confid, formula, versus, connect, discret, man

The hierarchical nature of the model allows counting the subgroups for each group (table 8) but the results are not interesting. The numerosity of the subgroups does not appear to be linked neither with the more applied fields, nor with the time in which a group reached its peak.

It is time to take a step back, to lose the model from sight and to try to refrain the question in a more concrete way. Pluralism is the coexistence of different programmes, each with a different goal and, it is reasonable to assume, slightly different lexicons. So, in a pluralistic period the number of different words used should be higher than in a homogeneous period.

To illustrate this idea, the number of unique words used in each year is measured, and then it is compared to the number of papers in the sample and to the total words used (i.e. the total length of the cleaned papers published in a year). Figure 5 summarizes the comparison.

The number of unique words used grows from the beginning of the sample and since the eighties it grows faster than the number of newly published documents. This means that the probability that a new document introduces a new word (for example because a new subfield has reached the status to be accepted in the top journals) is increasing too.

Table 7: The composition of each group as a mixture of topics. Only the topics which represent at least 10% of the group are listed

Applied Micro	0.29*Applied Micro 1 + 0.15*Applied Micro 2 + ...
Econometrics	0.42*Applied Micro 1 + 0.14*Credit + ...
Game Theory - Theory	0.29*Applied Micro 1 + 0.17*Maths + ...
Game Theory - Auctions	0.28*Applied Micro 1 + 0.12*Maths + ...
Game Theory - Elections	0.33*Applied Micro 1 + ...
Industrial Organization	0.26*Applied Micro 1 + ...
Labour	0.28*Applied Micro 1 + 0.19*Applied Micro 2 + ...
Macro 1	0.25*Applied Micro 1 + 0.16*Labour + ...
Macro 2	0.25*Applied Micro 1 + 0.14*Labour + 0.11*Credit + ...
Macro Econometrics	0.30*Applied Micro 1 + 0.12*Maths + ...
Trade	0.23*Applied Micro 1 + 0.13*Applied Micro 2 + ...

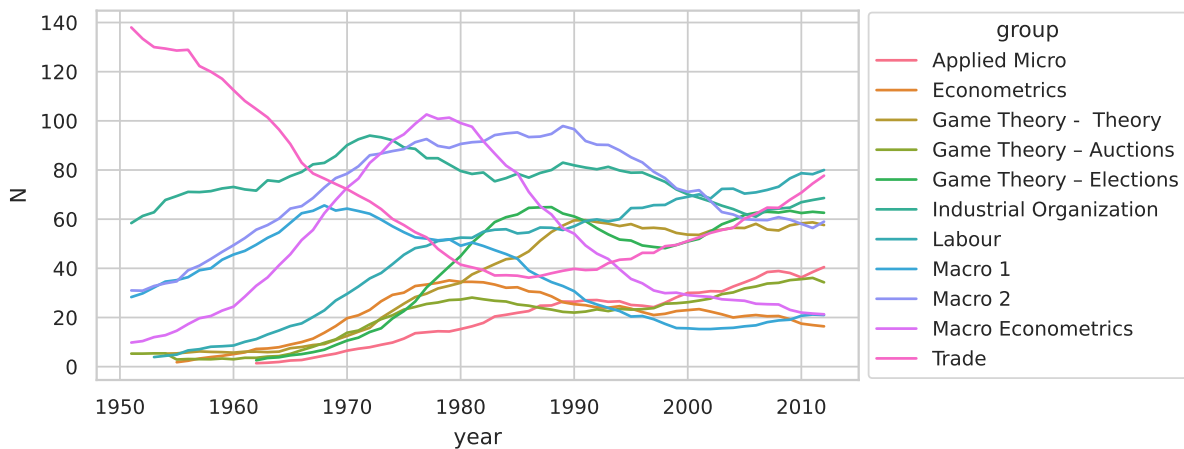


Figure 3: Number of documents in each group during time. Smoothed with a 10-year rolling average

On the other hand, the total number of words published is increasing way faster than the number of unique words, because of the progressive increment in length of the papers. This suggests that the increased lexical variety is not a matter of diversification, but a necessary by-product of longer articles.

Using these data, I am not able to see a way to exclude one of the two explanations.

To summarize, I have not found in this topic model a good instrument to measure the (eventual) advent of a new mainstream pluralism. This can be the consequence of the specific algorithm used or a signal that the research question has to be rephrased in a more theory-aware way.

9. Limitations

This study is an early work in its fields, and probably would reach a mature stage only with a further effort.

A better source of data (maybe with an agreement with some commercial providers) could improve the results, particularly extending the time period could highlight an eventual process of specialization stemmed from the 2008 crisis, which could be interpreted as an effect of mainstream pluralism.

The idea of focusing on the top journals as a proxy for mainstream convinces me. But which are the top journals is debatable. If only a smaller time period is going to be considered, approximately from the eighties until today, the Top5 can be an exhaustive group. But the

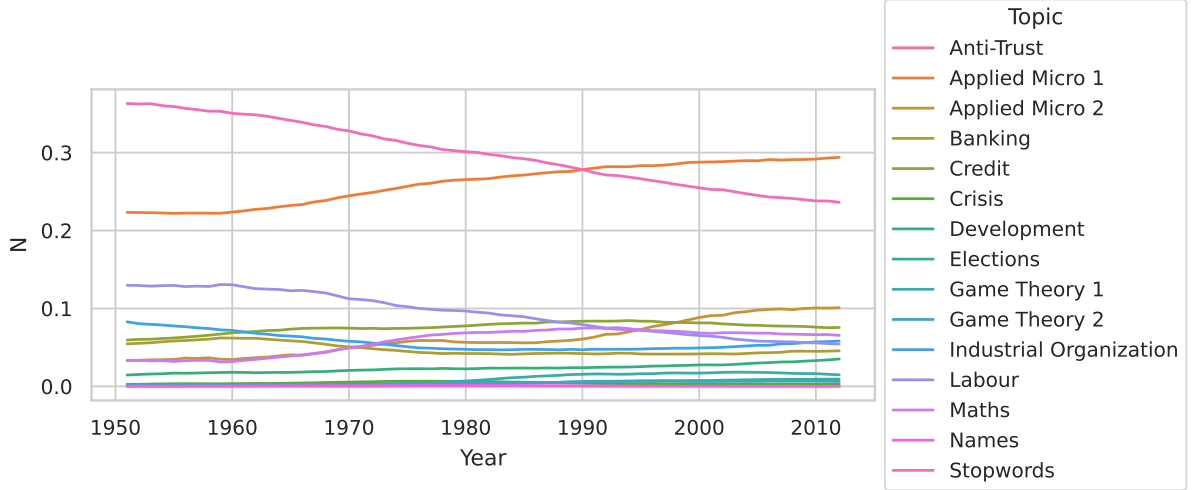


Figure 4: Prevalence of each topic during time. Smoothed with a 10-year rolling average

Table 8: The number of groups in the more detailed level of the hierarchy for each group in the level of interest

Industrial Organization	10
Game Theory – Auctions	8
Applied Micro	6
Macro 1	6
Macro 2	6
Macro Econometrics	6
Game Theory - Theory	5
Game Theory – Elections	5
Labour	5
Trade	4
Econometrics	2

equilibria in the discipline in the previous decades were less defined. Moreover, the very choice to use textual data rather than, for example, bibliometric ones is not written in stone.

There are at least two steps which could benefit from a robustness check: the cleaning of the dataset and the inference of the model.

The choice of stemming the dataset is arbitrary, and it should be checked if the alternatives (lemmatization, embedding and doing nothing) impact the model inferred. Similarly, the strategy used to reduce the size of the vocabulary, cutting the words present in many or few documents, is only one of the possibility, and maybe not the best one (Gerlach et al., 2019).

Allowing that inferring a topic model is the right way to approach this problem, there is not a consensus in literature about which algorithm performs better and, maybe more importantly, how to compare them (a recent and rare attempt is Shi et al., 2019). I still believe that the limitations of LDA, particularly the need to fix a priori the number of topics, requires pivoting to some more modern algorithms. Which is still an open question.

Finally, the labels given to the topics and the groups are subjective, and the method chosen is dictated more by the data at disposition than by some fine reasoning. JSTOR is a fantastic resource to get the word count of many (but not all!) journals¹⁹, but it does not keep track of the citations received by a work, or the JEL codes attributed to it. Moreover, there is not

¹⁹But it does not provide the full text, which can be a useful resource for more advanced machine learning models.

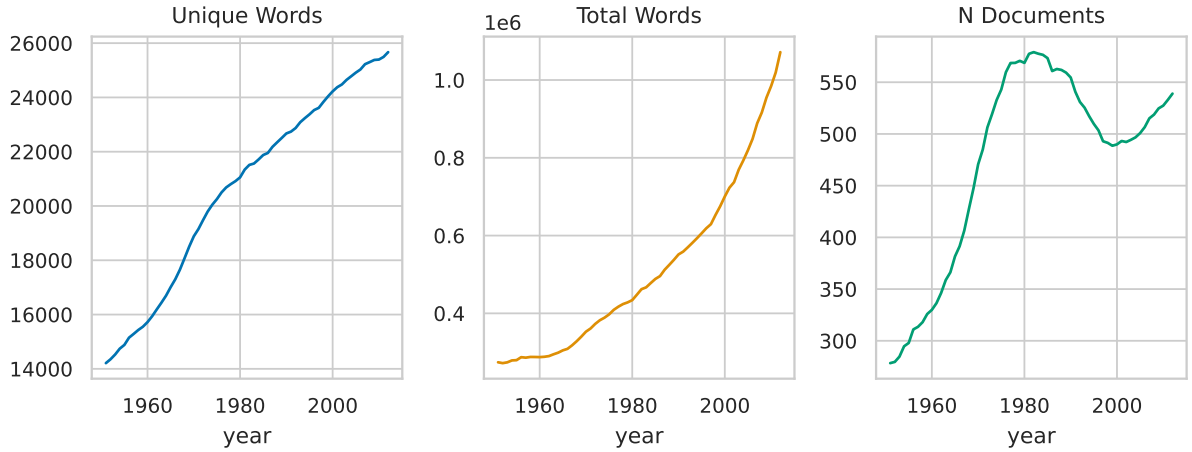


Figure 5: From left to right: number of unique words per year in the sample; total number of words per year in the sample; number of documents per year in the sample

a public database that relates the JSTOR identifier, with the Scopus one²⁰ or the EconLit one²¹, making the operation of mixing metadata from different sources really difficult and time-consuming. Two other strategies to interpret the groups have been discarded for this reason: label each group with the most frequent JEL codes, and characterize each group with the most cited documents in it.

10. What's next?

I can suggest four directions in which this work can continue.

The first one is to explore another finer-grained level of the hierarchy to look for specific subfields described in literature as part of the mainstream pluralism (Davis, 2006, §1), and observe their evolution through time.

The second one is to substitute hSBM with another algorithm (maybe BERTopic), and observe what changes and what not.

The third one is to enhance the JSTOR database with citations data (from Scopus) and the JEL codes of the articles (from EconLit). Doing so requires to limit the time span (going back no later than 1980) and a lot of time (and probably exploited labour). The dataset obtained should permit to tackle the problem from multiple angles at a time, for example generalizing to a communities' detection problem on a multi-layered graph, or exploring the correlation between different ways to signal the belonging to a research community.

The last one is probably the most important and most radical: to forget about this work, doing a deeper qualitative study to highlight the fundamental characteristics of mainstream pluralism and to translate them into quantitative questions. There is a debate in the part of the digital humanities' community who studies history of ideas: is it better to use complex models to highlight some general features that are not observable by a human's eye (like in this work), or is it better to formulate more precise questions and try to measure some very specific features (like in Babbioni and Ciruzzi (2022))? I am inclined to the second approach, mostly because the data manipulated in this kind of studies are rarely high quality data, and more complex algorithms

²⁰A lot of the older document are missing from Scopus. As consequence, every study that relies on it to get bibliometric data, even just to identify the most relevant paper in a cluster, is constrained in the time period that can be studied.

²¹The EconLit database associates to every record its JEL codes. But there are some different taxonomies of JEL codes in different periods. The last big update happened in 1990 (Cherrier, 2017), without a clear way to convert the previous taxonomy in the new one.

are able to find “features” even when only noise is present. The possible side effects of simpler and understandable measures are easier to spot, allowing for a more conscious management of the noise in the data (like a poor OCR, or an imperfect coupling between two data sources).

455 Unfortunately, at the time of writing, I have not in mind a characterization of the mainstream pluralism (or alternatively of what makes a research programme “post-neoclassical” (Redigonda, 2023)) clear enough to hypothesize some measurable proxies for it. But it can be the topic of another paper.

References

- Ambrosino, A., Cedrini, M., Davis, J. B., Fiori, S., Guerzoni, M., & Nuccio, M. (2018). What topic modeling could reveal about the evolution of economics. *Journal of Economic Methodology*, 25(4), 329–348. <https://doi.org/10/gf359h>
- Angelov, D. (2020, August 19). *Top2Vec: Distributed Representations of Topics*. Retrieved September 25, 2021, from <http://arxiv.org/abs/2008.09470>
- Angrist, J., Azoulay, P., Ellison, G., Hill, R., & Lu, S. F. (2017). Economic Research Evolves: Fields and Styles. *American Economic Review*, 107(5), 293–297. <https://doi.org/10/gdgx8x>
- Babbiotti, P., & Ciruzzi, M. (2022). Doing philosophy as opening parentheses: Quantifying the use of parentheses in Stanley Cavell’s style. *Inquiry*, 0(0), 1–28. <https://doi.org/10.1080/0020174X.2022.2131621>
- Backhouse, R., & Biddle, J. (2000). The Concept of Applied Economics: A History of Ambiguity and Multiple Meanings. *History of Political Economy*, 32, 1–24. https://doi.org/10.1215/00182702-32-Suppl_1-1
- Backhouse, R., & Cherrier, B. (2016, October 1). ‘It’s Computerization, Stupid!’ *The Spread of Computers and the Changing Roles of Theoretical and Applied Economics* (SSRN Scholarly Paper No. ID 2781253). Social Science Research Network. Rochester, NY. <https://doi.org/10.2139/ssrn.2781253>
- Backhouse, R., & Cherrier, B. (2017). The Age of the Applied Economist: The Transformation of Economics since the 1970s. *History of Political Economy*, 49, 1–33. <https://doi.org/10.1215/00182702-4166239>
- Backhouse, R., Middleton, R., & Tribe, K. (1997). ‘Economics Is What Economists Do’, *But What Do the Numbers Tell Us?* <https://doi.org/10.2139/ssrn.2942733>
- Billings, B. B., & Viksnins, G. J. (1972). THE RELATIVE QUALITY OF ECONOMICS JOURNALS: AN ALTERNATIVE RATING SYSTEM. *Economic Inquiry*, 10(4), 467–469. <https://doi.org/10.1111/j.1465-7295.1972.tb01568.x>
- Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent dirichlet allocation. *Journal of Machine Learning Research*, 3, 993–1022.
- Campello, R. J. G. B., Moulavi, D., & Sander, J. (2013). Density-Based Clustering Based on Hierarchical Density Estimates. In J. Pei, V. S. Tseng, L. Cao, H. Motoda & G. Xu (Eds.). D. Hutchison, T. Kanade, J. Kittler, J. M. Kleinberg, F. Mattern, J. C. Mitchell, M. Naor, O. Nierstrasz, C. Pandu Rangan, B. Steffen, M. Sudan, D. Terzopoulos, D. Tygar, M. Y. Vardi & G. Weikum (Red.), *Advances in Knowledge Discovery and Data Mining* (pp. 160–172, Vol. 7819). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-37456-2_14
- Card, D., & DellaVigna, S. (2013). Nine Facts about Top Journals in Economics. *Journal of Economic Literature*, 51(1), 144–161. <https://doi.org/10.1257/jel.51.1.144>
- Cedrini, M., & Fontana, M. (2018). Just another niche in the wall? How specialization is changing the face of mainstream economics. *Cambridge Journal of Economics*, 42(2), 427–451. <https://doi.org/10/gfxn7t>
- Cherrier, B. (2017). Classifying Economics: A History of the JEL Codes. *Journal of Economic Literature*, 55(2), 545–579. <https://doi.org/10/gdh6pn>
- Cherrier, B., & Svorenčík, A. (2018). The quantitative turn in the history of economics: Promises, perils and challenges. *Journal of Economic Methodology*, 25(4), 367–377. <https://doi.org/10.1080/1350178X.2018.1529217>
- Claveau, F., & Gingras, Y. (2016). Macrodynamics of Economics: A Bibliometric History. *History of Political Economy*, 48(4), 551–592. <https://doi.org/10/gf4g58>
- Coats, A. W. (1971). The Role of Scholarly Journals in the History of Economics: An Essay. *Journal of Economic Literature*, 9(1), 29–44. Retrieved October 8, 2022, from <https://www.jstor.org/stable/2720804>

- Davis, J. B. (2006). The turn in economics: Neoclassical dominance to mainstream pluralism? *Journal of Institutional Economics*, 2(1), 1–20. <https://doi.org/10/bzn5sm>
- Davis, J. B. (2019a, January 28). *Economics and Economic Methodology in a Core-Periphery Economic World*. <https://doi.org/10.2139/ssrn.3324569>
- Davis, J. B. (2019b). Specialization, fragmentation, and pluralism in economics. *The European Journal of the History of Economic Thought*, 26(2), 271–293. <https://doi.org/10.1080/09672567.2018.1555604>
- Dusansky, R., & Vernon, C. J. (1998). Rankings of U.S. Economics Departments. *Journal of Economic Perspectives*, 12(1), 157–170. <https://doi.org/10.1257/jep.12.1.157>
- Edwards, J., Giraud, Y., & Schinckus, C. (2018). A quantitative turn in the historiography of economics? *Journal of Economic Methodology*, 25(4), 283–290. <https://doi.org/10.1080/1350178X.2018.1529133>
- Ellison, G. (2003, February 27). *The Slowdown of the Economics Publishing Process*. Retrieved February 6, 2023, from <https://papers.ssrn.com/abstract=327004>
- Fontana, M., Montobbio, F., & Racca, P. (2019). TOPICS AND GEOGRAPHICAL DIFFUSION OF KNOWLEDGE IN TOP ECONOMIC JOURNALS. *Economic Inquiry*, 57(4), 1771–1797. <https://doi.org/10/ggz77h>
- Fourcade, M., Ollion, E., & Algan, Y. (2015). The Superiority of Economists. *Journal of Economic Perspectives*, 29(1), 89–114. <https://doi.org/10.1257/jep.29.1.89>
- Galbraith, J. K. (1994). *A short history of financial euphoria*. Whittle Books in association with Penguin Books.
- Gerlach, M., Peixoto, T. P., & Altmann, E. G. (2018). A network approach to topic models. *Science Advances*, 4(7), eaaq1360. <https://doi.org/10/gdxnxq>
- Gerlach, M., Shi, H., & Amaral, L. A. N. (2019). A universal information theoretic approach to the identification of stopwords. *Nature Machine Intelligence*, 1(12), 606–612. <https://doi.org/10.1038/s42256-019-0112-6>
- Griffiths, T. L., & Steyvers, M. (2004). Finding scientific topics. *Proceedings of the National Academy of Sciences*, 101, 5228–5235. <https://doi.org/10.1073/pnas.0307752101>
- Grootendorst, M. (2022, March 11). *BERTopic: Neural topic modeling with a class-based TF-IDF procedure*. arXiv: 2203.05794 [cs]. <https://doi.org/10.48550/arXiv.2203.05794>
- Hamermesh, D. S. (2013). Six Decades of Top Economics Publishing: Who and How? *Journal of Economic Literature*, 51(1), 162–172. <https://doi.org/10.1257/jel.51.1.162>
- Hawkins, R. G., Ritter, L. S., & Walter, I. (1973). What Economists Think of Their Journals. *Journal of Political Economy*, 81(4), 1017–1032. <https://doi.org/10.1086/260098>
- Heck, J. L., & Zaleski, P. A. (2006). The most frequent contributors to the elite economics journals: Half century of contributions to the “Blue ribbon eight”. *Journal of Economics and Finance*, 30(1), 1–37. <https://doi.org/10/drboxd5>
- Heckman, J. J., & Moktan, S. (2020). Publishing and Promotion in Economics: The Tyranny of the Top Five. *Journal of Economic Literature*, 58(2), 419–470. <https://doi.org/10/ggzndw>
- Hofmann, T. (1999). *Probabilistic Latent Semantic Analysis*. Retrieved June 2, 2020, from <http://arxiv.org/abs/1301.6705>
- Kelly, M. A., & Bruestle, S. (2011). TREND OF SUBJECTS PUBLISHED IN ECONOMICS JOURNALS 1969-2007. *Economic Inquiry*, 49(3), 658–673. <https://doi.org/10/cgw9zz>
- Kim, E. H., Morse, A., & Zingales, L. (2006). What Has Mattered to Economics Since 1970. *Journal of Economic Perspectives*, 20(4), 189–202. <https://doi.org/10.1257/jep.20.4.189>
- Kim, E. H., Morse, A., & Zingales, L. (2009). Are elite universities losing their competitive edge? *Journal of Financial Economics*, 93(3), 353–381. <https://doi.org/10.1016/j.jfineco.2008.09.007>

- Kim, J., & Park, H. (2008). *Sparse Nonnegative Matrix Factorization for Clustering* (Technical Report). Georgia Institute of Technology. Retrieved September 29, 2021, from <https://smartech.gatech.edu/handle/1853/20058>
- Kosnik, L.-R. (2018). A SURVEY OF JEL CODES: WHAT DO THEY MEAN AND ARE THEY USED CONSISTENTLY?: A SURVEY OF JEL CODES. *Journal of Economic Surveys*, 32(1), 249–272. <https://doi.org/10/gf4g6n>
- Lancichinetti, A., Siler, M. I., Wang, J. X., Acuna, D., K rding, K., & Amaral, L. A. N. (2015). High-Reproducibility and High-Accuracy Method for Automated Topic Classification. *Physical Review X*, 5(1), 011007. <https://doi.org/10/gfrbjc>
- Liebowitz, S. J., & Palmer, J. P. (1984). Assessing the Relative Impacts of Economics Journals. *Journal of Economic Literature*, 22(1), 77–88. <http://www.jstor.org/stable/2725228>
- Malouin, J.-L., & -Francois Outreville, J. (1987). The relative impact of economics journals: A cross-country survey and comparison. *Journal of Economics and Business*, 39(3), 267–277. [https://doi.org/10.1016/0148-6195\(87\)90022-1](https://doi.org/10.1016/0148-6195(87)90022-1)
- Marchionatti, R., & Cedrini, M. (2017). *Economics as social science: Economics imperialism and the challenge of interdisciplinary*. Routledge, Taylor & Francis Group.
- McInnes, L., Healy, J., & Astels, S. (2017). Hdbscan: Hierarchical density based clustering. *Journal of Open Source Software*, 2(11), 205. <https://doi.org/10.21105/joss.00205>
- McInnes, L., Healy, J., & Melville, J. (2020, September 17). *UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction*. Retrieved September 25, 2021, from <http://arxiv.org/abs/1802.03426>
- Montesinos, H., & Brice, B. (2019, November 25). *The Era of Evidence*.
- Moore, W. J. (1972). THE RELATIVE QUALITY OF ECONOMICS JOURNALS: A SUGGESTED RATING SYSTEM. *Economic Inquiry*, 10(2), 156–169. <https://doi.org/10.1111/j.1465-7295.1972.tb01596.x>
- Moretti, F. (2013). *Distant reading*. Verso.
- Peixoto, T. P. (2019, November 23). *Bayesian stochastic blockmodeling*. <https://doi.org/10.1002/9781119483298.ch11>
- Redigonda, M. (2023). On complexity as meta-theory: A perspective from economics. <https://doi.org/10.31219/osf.io/hnvzx>
- Shi, H., Gerlach, M., Diersen, I., Downey, D., & Amaral, L. A. N. (2019, January 28). *A new evaluation framework for topic modeling algorithms based on synthetic corpora*. Retrieved February 27, 2020, from <http://arxiv.org/abs/1901.09848>
- Teh, Y., Jordan, M., Beal, M., & Blei, D. (2005). Sharing Clusters among Related Groups: Hierarchical Dirichlet Processes. *Advances in Neural Information Processing Systems*, 17. Retrieved September 24, 2021, from <https://papers.nips.cc/paper/2004/hash/fb4ab556bc42d6f0ee0f9e24ec4d1af0-Abstract.html>

A. Additional tables

Table 9: The five papers most representative of each group, measured by the cosine similarity with the word count in the topic as whole

Applied Micro	Household Location and Race: Estimates of a Multinomial Logit Model	Stuart A. Gabriel, Stuart S. Rosenthal	1989	The Review of Economics and Statistics
Applied Micro	The Macroeconomic Implications of Rising Wage Inequality in the United States	Jonathan Heathcote, Kjetil Storesletten, Giovanni L. Violante	2010	Journal of Political Economy
Applied Micro	Relative Wage Movements and the Distribution of Consumption	Orazio Attanasio, Steven J. Davis	1996	Journal of Political Economy
Applied Micro	Efficiency and Equity in the Provision of Public Education	R. Hamilton Lankford	1985	The Review of Economics and Statistics
Applied Micro	Alimony Rights and Intrahousehold Allocation of Resources: Evidence from Brazil	Marcos A. Rangel	2006	The Economic Journal
Econometrics	Asymptotic Normality, When Regressors Have a Unit Root	Kenneth D. West	1988	Econometrica
Econometrics	Probit with Dependent Observations	Dale J. Poirier, Paul A. Ruud	1988	The Review of Economic Studies
Econometrics	Large Sample Properties of Generalized Method of Moments Estimators	Lars Peter Hansen	1982	Econometrica
Econometrics	TESTING FOR COMMON CONDITIONALLY HETEROSKEDASTIC FACTORS	Prosper Dovonon, Eric Renault	2013	Econometrica
Econometrics	Multiple Time Series Regression with Integrated Processes	P. C. B. Phillips, S. N. Durlauf	1986	The Review of Economic Studies
Game Theory - Theory	EFFICIENCY IN GAMES WITH MARKOVIAN PRIVATE INFORMATION	Juan F. Escobar, Juuso Toikka	2013	Econometrica
Game Theory - Theory	Global Games and Equilibrium Selection	Hans Carlsson, Eric van Damme	1993	Econometrica
Game Theory - Theory	Rational Expectations in Games	Robert J. Aumann, Jacques H. Dreze	2008	The American Economic Review
Game Theory - Theory	Endogenous Games and Mechanisms: Side Payments among Players	Matthew O. Jackson, Simon Wilkie	2005	The Review of Economic Studies
Game Theory - Theory	COOPERATION IN STRATEGIC GAMES REVISITED	Adam Kalai, Ehud Kalai	2013	The Quarterly Journal of Economics
Game Theory - Auctions	Standard Auctions with Financially Constrained Bidders	Yeon-Koo Che, Ian Gale	1998	The Review of Economic Studies

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Table 9: The five papers most representative of each group, measured by the cosine similarity with the word count in the topic as whole

Game Theory – Auctions	The Relevance of a Choice of Auction Format in a Competitive Environment	Matthew O. Jackson, Ilan Kremer	2006	The Review of Economic Studies
Game Theory – Auctions	Private Information and Altruism in Bidding Roscas	Stefan Klonner	2008	The Economic Journal
Game Theory – Auctions	Bidding with Securities: Auctions and Security Design	Peter M. DeMarzo, Ilan Kremer, Andrzej Skrzypacz	2005	The American Economic Review
Game Theory – Auctions	Discrete Bids and Empirical Inference in Divisible Good Auctions	JAKUB KASTL	2011	The Review of Economic Studies
Game Theory – Elections	On the Theory of Strategic Voting	David P. Myatt	2007	The Review of Economic Studies
Game Theory – Elections	Industrial Economics: An Overview	Richard Schmalensee	1988	The Economic Journal
Game Theory – Elections	Why Democracies Produce Efficient Results	Donald Wittman	1989	Journal of Political Economy
Game Theory – Elections	Misconceptions and Political Outcomes	David Romer	2003	The Economic Journal
Game Theory – Elections	Vote Buying: General Elections	Eddie Dekel, Matthew O. Jackson, Asher Wolinsky	2008	Journal of Political Economy
Industrial Organization	The Economics of Development: A Survey	Nicholas Stern	1989	The Economic Journal
Industrial Organization	The Interindustry Relations Study for 1947	W. Duane Evans, Marvin Hoffenberg	1952	The Review of Economics and Statistics
Industrial Organization	The British Labor Party and Domestic Reform	Mary E. Murphy	1946	Journal of Political Economy
Industrial Organization	Domestic Trade in an Underdeveloped Country–Turkey	William H. Nicholls	1951	Journal of Political Economy
Industrial Organization	The Economics of the Next Century	Austin Robinson	1991	The Economic Journal
Labour	Wage Risk and Employment Risk over the Life Cycle	Hamish Low, Costas Meghir, Luigi Pistaferri	2010	The American Economic Review
Labour	On-the-Job Search and Precautionary Savings	JEREMY LISE	2013	The Review of Economic Studies
Labour	MODELING EARNINGS DYNAMICS	Joseph G. Altonji, Anthony A. Smith, Jr., Ivan Vidangos	2013	Econometrica
Labour	Employer-Employee Interaction and the Duration of Unemployment	John T. Warner, J. Carl Poindexter, Jr., Robert M. Fearn	1980	The Quarterly Journal of Economics
Labour	Factors Affecting the Output and Quit Propensities of Production Workers	Roger Klein, Richard Spady, Andrew Weiss	1991	The Review of Economic Studies

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Table 9: The five papers most representative of each group, measured by the cosine similarity with the word count in the topic as whole

Macro 1	The Problem of Economic Instability	Emile Despres, Albert G. Hart, Milton Friedman, Paul A. Samuelson, Donald H. Wallace	1950	The American Economic Review
Macro 1	Recent Developments in Macroeconomics	Stanley Fischer	1988	The Economic Journal
Macro 1	Why Real Interest Rates Were So Low in the 1970s	James A. Wilcox	1983	The American Economic Review
Macro 1	The Monetarist Controversy or, Should We Forsake Stabilization Policies?	Franco Modigliani	1977	The American Economic Review
Macro 1	The History of the Phillips Curve: Consensus and Bifurcation	Robert J. Gordon	2011	Economica
Macro 2	Recent Developments in Macroeconomics	Stanley Fischer	1988	The Economic Journal
Macro 2	The Conduct of Monetary Policy	Charles Goodhart	1989	The Economic Journal
Macro 2	The Problem of Economic Instability	Emile Despres, Albert G. Hart, Milton Friedman, Paul A. Samuelson, Donald H. Wallace	1950	The American Economic Review
Macro 2	Exchange Rate Economics: 1986	Rudiger Dornbusch	1987	The Economic Journal
Macro 2	International Real Business Cycles	David K. Backus, Patrick J. Kehoe, Finn E. Kydland	1992	Journal of Political Economy
Macro Econometrics	Recent Developments in Macroeconomics	Stanley Fischer	1988	The Economic Journal
Macro Econometrics	The Monetarist Controversy or, Should We Forsake Stabilization Policies?	Franco Modigliani	1977	The American Economic Review
Macro Econometrics	When Demand Creates its Own Supply: Saving Traps	CHRISTOPHE CHAMLEY	2014	The Review of Economic Studies
Macro Econometrics	FISCAL POLICY IN A TRACTABLE LIQUIDITY-CONSTRAINED ECONOMY	Edouard Challe, Xavier Ragot	2011	The Economic Journal
Macro Econometrics	Behavioral Macroeconomics and Macroeconomic Behavior	George A. Akerlof	2002	The American Economic Review
Trade	Household and Economy: Toward a New Theory of Population and Economic Growth	Marc Nerlove	1974	Journal of Political Economy
Trade	The Economic Approach to Social Questions	Harry G. Johnson	1968	Economica
Trade	The Economics of Development: A Survey	Nicholas Stern	1989	The Economic Journal
Trade	Alimony Rights and Intrahousehold Allocation of Resources: Evidence from Brazil	Marcos A. Rangel	2006	The Economic Journal

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Table 9: The five papers most representative of each group, measured by the cosine similarity with the word count in the topic as whole

Trade	The British Labor Party and Domestic Reform	Mary E. Murphy	1946	Journal of Political Economy
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B. Graph and Groups for the other seeds

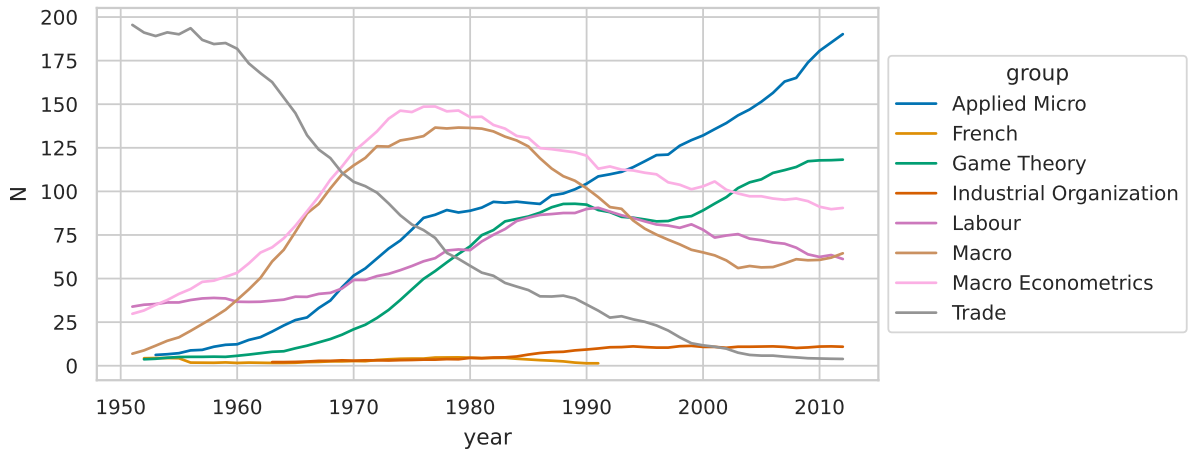


Figure 6: Number of documents in each group during time. Smoothed with a 10-year rolling average. Seed 1000

Table 10: The documents-group and the twenty most frequent words in each of them. Seed 1000

Applied Micro	household, educ, women, job, children, health, student, unemploy, insur, hous, hour, panel, child, citi, dummi, black, men, male, parent, femal
French	des, les, que, est, par, dan, pour, une, qui, nous, bien, prix, sont, sur, pas, cett, fonction, tout, hedg, peut
Game Theory	game, player, payoff, bid, auction, lemma, buyer, seller, belief, bidder, equilibria, signal, match, bargain, finit, nash, princip, vote, experiment, pair
Industrial Organization	tariff, export, plant, bargain, debt, shock, foreign, union, domest, entri, manufactur, agreement, game, job, labour, volatil, exit, regim, instrument, panel
Labour	unemploy, job, household, union, labour, match, hous, bargain, insur, hour, educ, compens, compani, vote, shock, employe, rent, skill, parti, corpor
Macro	inflat, shock, export, foreign, forecast, domest, unemploy, debt, household, bond, nomin, credit, manufactur, liquid, cycl, currenc, portfolio, labour, investor, corpor
Macro Econometrics	shock, inflat, matrix, export, lemma, foreign, debt, asymptot, converg, tariff, path, forecast, domest, credit, volatil, bond, investor, borrow, finit, portfolio
Trade	export, foreign, union, agricultur, war, manufactur, credit, inflat, domest, million, capac, corpor, compani, debt, unemploy, professor, volum, farm, soviet, plant

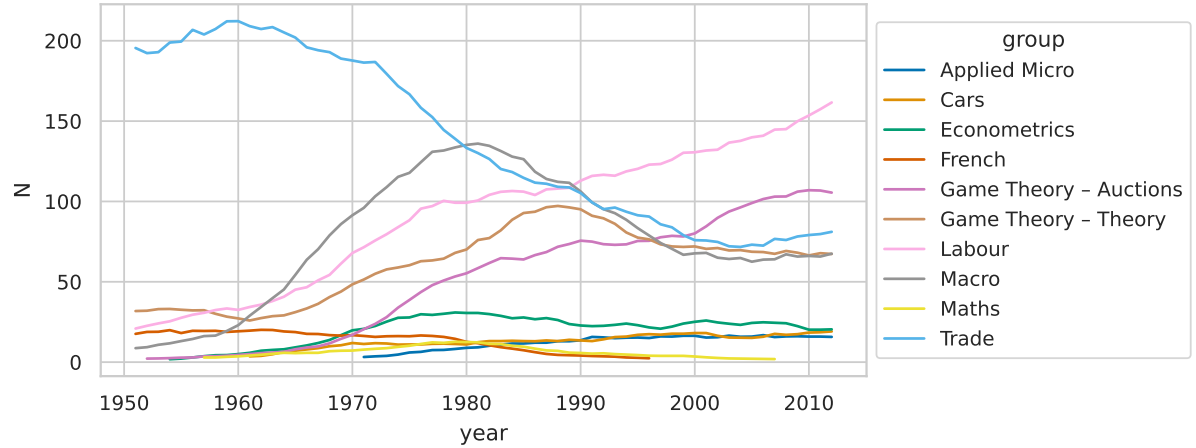


Figure 7: Number of documents in each group during time. Smoothed with a 10-year rolling average.
Seed 1002

Table 11: The documents-group and the twenty most frequent words in each of them. Seed 1002

Applied Micro	crime, educ, vote, insur, elect, household, parti, debt, borrow, player, team, sharehold, loan, credit, monitor, unemploy, taxat, labour, investor, citi
Cars	car, shock, emiss, manufactur, vehicl, plant, fuel, pollut, hour, inventori, household, energi, hous, inflat, retail, regul, electr, season, environment, panel
Econometrics	asymptot, matrix, lemma, moment, likelihood, covari, converg, densiti, instrument, finit, identif, uniform, nonparametr, simul, null, root, sequenc, nonlinear, bootstrap, regressor
French	des, les, que, cycl, labour, est, capac, par, dan, pour, une, path, acceler, tariff, optimum, qui, accumul, durabl, nous, monopolist
Game Theory – Auctions	game, player, payoff, bid, auction, vote, lemma, buyer, belief, signal, match, seller, equilibria, bidder, voter, parti, experiment, learn, bargain, round
Game Theory – Theory	game, debt, equilibria, player, payoff, buyer, lemma, bargain, tariff, borrow, seller, credit, labour, unemploy, shock, surplus, foreign, loan, union, investor
Labour	household, educ, unemploy, job, children, women, health, student, insur, hour, hous, child, parent, men, panel, male, black, labour, femal, occup
Macro	inflat, shock, forecast, export, nomin, foreign, volatil, bond, household, domest, portfolio, matrix, path, cycl, unemploy, debt, investor, labour, longrun, panel
Maths	convex, lemma, matrix, concav, sequenc, finit, axiom, equilibria, monoton, path, feasibl, converg, endow, labour, homogen, algorithm, map, nonneg, mathemat, tenant
Trade	export, foreign, agricultur, manufactur, union, domest, credit, war, land, plant, inflat, corpor, compani, debt, tariff, patent, innov, farm, capac, loan

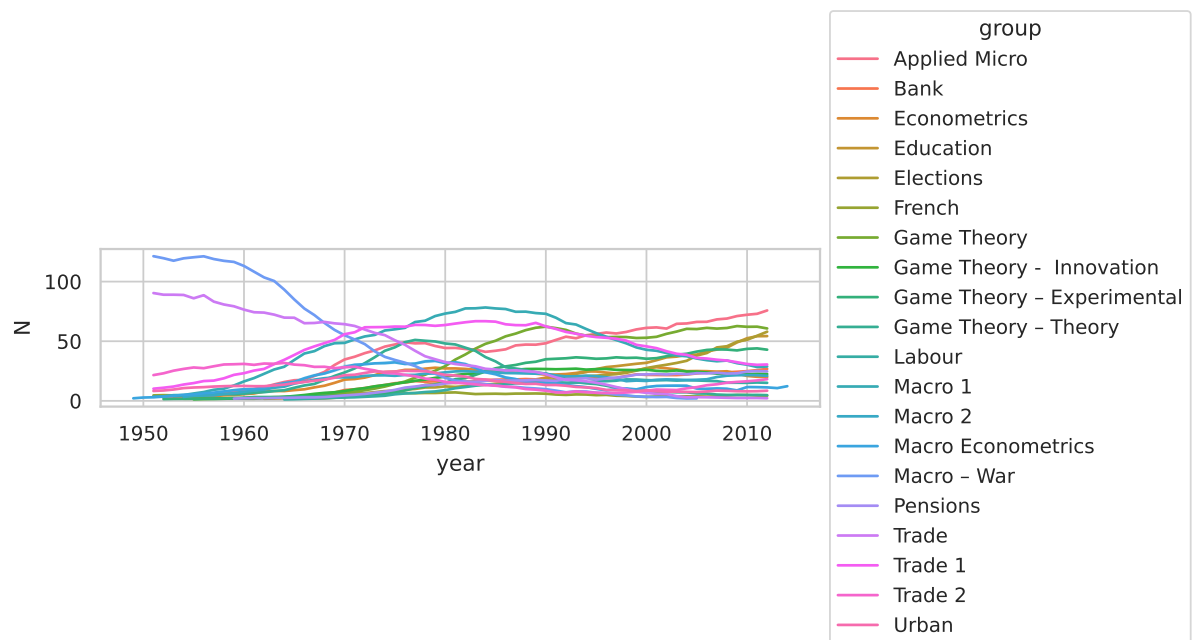


Figure 8: Number of documents in each group during time. Smoothed with a 10-year rolling average.
Seed 1003

Table 12: The documents-group and the twenty most frequent words in each of them. Seed 1003

Applied Micro	educ, household, health, children, women, child, birth, parent, job, student, male, insur, men, immigr, unemploy, black, femal, fertil, white, occup
Bank	unemploy, loan, debt, credit, job, borrow, shock, household, match, equiti, bond, insur, investor, liquid, hous, portfolio, corpor, default, labour, mortgag
Econometrics	asymptot, matrix, lemma, moment, likelihood, converg, densiti, covari, instrument, finit, uniform, null, simul, identif, bootstrap, nonparametr, root, sequenc, nonlinear, confid
Education	student, educ, job, score, unemploy, panel, teacher, citi, yes, skill, dummi, women, children, hour, grade, household, plant, black, white, men
Elections	vote, elect, household, voter, parti, yes, dummi, panel, educ, candid, citi, day, hour, job, district, regul, crime, spend, car, vehicl
French	match, des, les, student, que, est, par, dan, pour, une, algorithm, qui, pair, nous, cycl, lemma, game, rank, roth, equilibria
Game Theory	player, game, payoff, buyer, signal, equilibria, seller, lemma, belief, princip, bid, auction, bargain, vote, parti, learn, entri, nash, finit, voter
Game Theory - Innovation	innov, entri, regul, bid, buyer, parti, seller, bargain, game, vote, union, surplus, network, capac, tariff, locat, monopoli, labour, compani, insur
Game Theory – Experimental	game, bid, auction, player, payoff, bidder, experiment, round, seller, buyer, bargain, you, belief, learn, win, signal, session, avers, cooper, job
Game Theory – Theory	game, lemma, axiom, convex, player, finit, sequenc, equilibria, subset, monoton, converg, pareto, coalit, feasibl, nash, core, pair, concav, payoff, represent
Labour	household, job, unemploy, educ, hour, match, insur, heterogen, simul, panel, retir, pension, instrument, labour, durat, hous, shock, unobserv, covari, health
Macro 1	inflat, shock, forecast, nomin, bond, foreign, volatil, domest, cycl, debt, currenc, unemploy, portfolio, credit, household, fiscal, macroeconom, longrun, liquid, dividend
Macro 2	shock, inflat, unemploy, forecast, volatil, cycl, inventori, debt, nomin, labour, panel, matrix, currenc, credit, manufactur, season, innov, foreign, regim, frequenc
Macro Econometrics	debt, shock, inflat, path, liquid, borrow, household, bond, lemma, matrix, labour, equilibria, portfolio, investor, credit, nomin, default, stochast, loan, converg
Macro Econometrics	tariff, domest, export, household, foreign, labour, budget, extern, matrix, optimum, taxat, subsidi, homogen, distort, path, protect, home, transform, slope, indiffer
Macro – War	export, union, foreign, war, credit, manufactur, domest, compani, soviet, debt, inflat, corpor, million, volum, agricultur, cycl, tariff, loan, capac, billion
Pensions	household, hous, educ, unemploy, retir, insur, hour, children, pension, labour, job, women, shock, child, panel, health, parent, cohort, bequest, home
Trade	export, agricultur, foreign, war, educ, professor, inflat, domest, transport, student, monopoli, manufactur, land, surplus, labour, million, cit, field, unemploy, man
Trade 1	export, manufactur, agricultur, tariff, foreign, patent, union, domest, household, land, plant, shock, farm, gdp, labour, unemploy, hour, innov, advertis, dummi
Trade 2	export, foreign, inflat, domest, tariff, unemploy, labour, manufactur, plant, skill, intens, bond, innov, entri, intermedi, affili, multin, credit, slope, transact
Urban	land, citi, hous, plant, transport, urban, agricultur, household, locat, rent, pollut, regul, farm, communiti, manufactur, union, capac, counti, hour, distanc

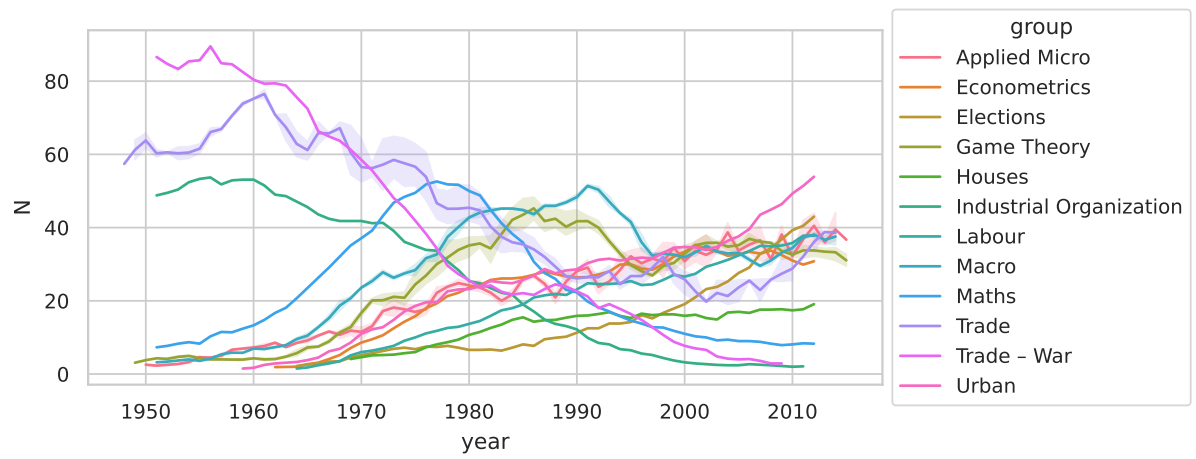


Figure 9: Number of documents in each group during time. Smoothed with a 10-year rolling average.
Seed 1004

Table 13: The documents-group and the twenty most frequent words in each of them. Seed 1004

Applied Micro	household, food, matrix, causal, record, perman, cloth, hous, budget, robbin, demograph, densiti, automobil, durabl, gini, tsls, ownership, unobserv, spend, fuel
Applied Micro	educ, household, health, children, women, student, child, job, birth, parent, insur, unemploy, men, immigr, score, black, femal, male, hour, dummi
Applied Micro	educ, white, black, women, hous, occup, male, farm, job, census, land, advertis, unemploy, citi, rent, men, agricultur, femal, household, migrat
Applied Micro	household, job, unemploy, educ, hour, retir, children, bequest, cohort, women, insur, parent, labour, child, durat, pension, panel, week, lifetim, male
Econometrics	asymptot, matrix, lemma, moment, covari, converg, likelihood, densiti, instrument, simul, identif, finit, null, uniform, nonparametr, panel, bootstrap, sequenc, root, nonlinear
Elections	vote, crime, elect, yes, student, citi, parti, voter, educ, panel, district, land, score, conflict, dummi, war, capita, agricultur, ethnic, spend
Game Theory	game, player, payoff, buyer, seller, lemma, equilibria, signal, belief, bargain, princip, trader, nash, finit, sequenc, learn, pair, convex, parti, sell
Game Theory	game, player, bid, auction, payoff, bidder, match, belief, signal, vote, buyer, equilibria, seller, voter, lemma, win, experiment, round, learn, parti
Game Theory	game, entri, bid, player, auction, unemploy, union, bargain, payoff, labour, monopoli, innov, regul, seller, buyer, parti, surplus, insur, extern, capac
Houses	credit, shock, hous, unemploy, inflat, loan, union, household, dividend, foreign, manufactur, equiti, panel, dummi, corpor, export, debt, borrow, volatil, portfolio
Industrial Organization	union, transport, unemploy, monopoli, compani, regul, capac, plant, hour, labour, manufactur, war, land, corpor, charg, commiss, rent, hous, entri, traffic
Labour	household, job, match, unemploy, insur, game, educ, labour, skill, health, player, avers, children, bargain, train, loan, experiment, employe, parent, her
Macro	inflat, shock, debt, bond, nomin, forecast, volatil, borrow, liquid, investor, credit, foreign, portfolio, domest, currenc, household, equiti, loan, default, unemploy
Macro	shock, inflat, forecast, cycl, matrix, household, spend, volatil, nomin, unemploy, hour, inventori, fiscal, instrument, disturb, longrun, macroeconom, fluctuat, residu, likelihood
Maths	lemma, matrix, avers, convex, finit, sequenc, des, path, axiom, les, portfolio, converg, concav, stochast, represent, equilibria, feasibl, que, monoton, endow
Trade	export, foreign, tariff, domest, labour, inflat, manufactur, unemploy, bond, capac, path, corpor, slope, intens, plant, household, optimum, debt, longrun, deprec
Trade	export, tariff, foreign, skill, domest, manufactur, patent, job, plant, innov, gdp, unemploy, panel, shock, labour, dummi, entri, union, educ, reform
Trade	export, inflat, foreign, credit, agricultur, domest, debt, manufactur, war, deposit, cycl, loan, unemploy, tariff, liquid, million, corpor, farm, billion, expans
Trade – War	export, foreign, union, agricultur, soviet, war, educ, million, farm, student, manufactur, gold, domest, compani, land, enterpris, currenc, credit, inflat, cit
Urban	citi, household, hous, land, locat, counti, plant, panel, patent, dummi, educ, car, innov, vehicl, rent, regul, electr, entri, urban, manufactur